

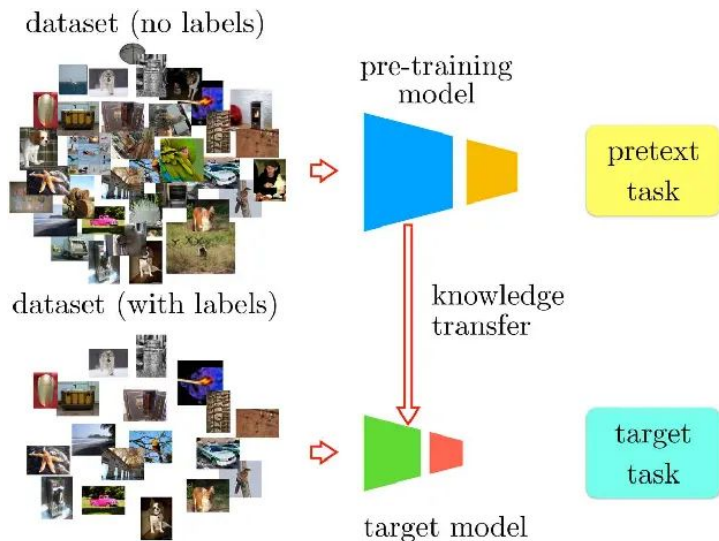
Representation Learning in Sensor DATA

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- PhD Candidate at IDI and DATA WP(September 2023)
 - Representation Learning in Sequential Data
- Supervisors:
 - Professors Kerstin Bach & Gavin Taylor

Self-supervised representation learning

Model trains itself to learn one part of the input from another part of the input through predictive or pretext learning



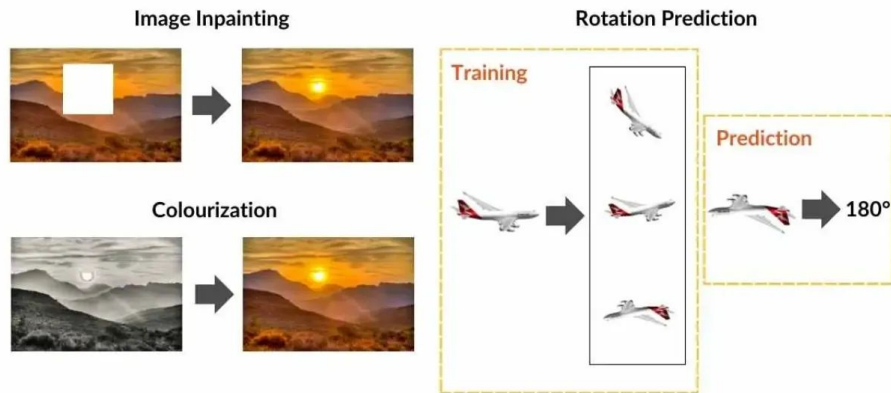
Pretext tasks

- Rotation
- Inpainting
- Colorization
- Jigsaw puzzle
- Reconstruction

Self-supervised representation learning

Model trains itself to learn one part of the input from another part of the input through predictive or pretext learning

Pretext Tasks

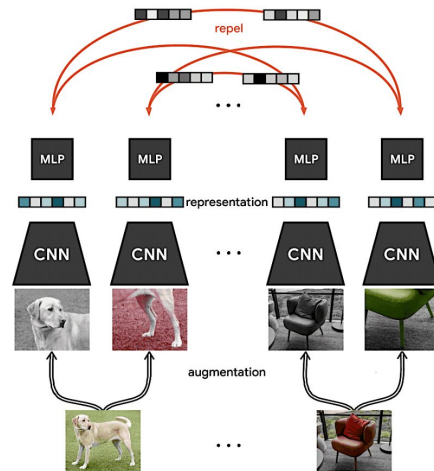
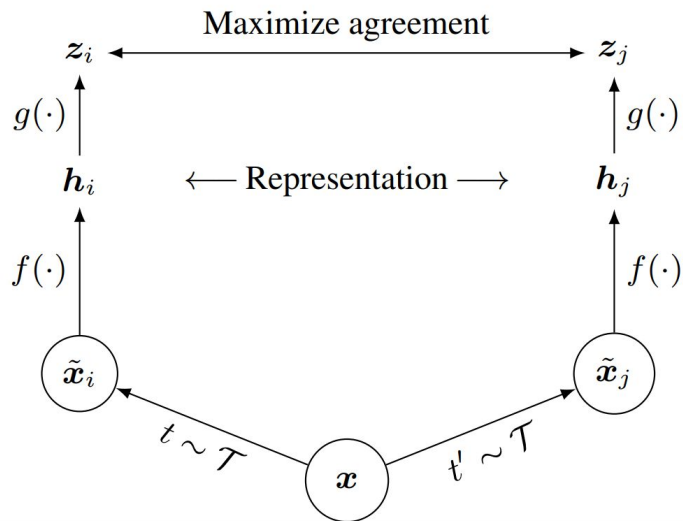


Pretext tasks

- Rotation
- Inpainting
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- Jigsaw puzzle
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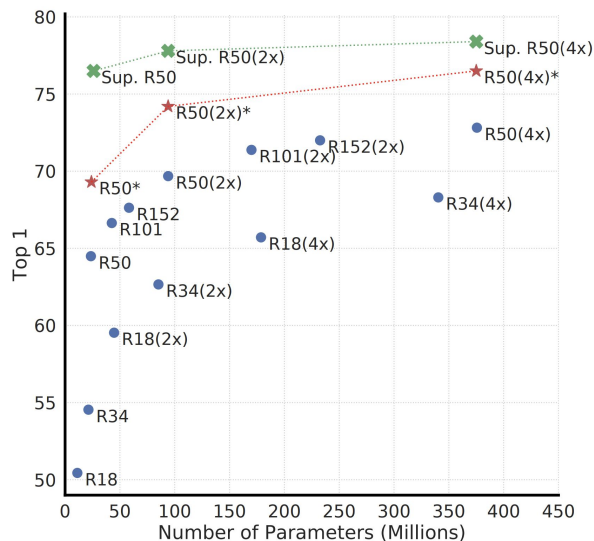
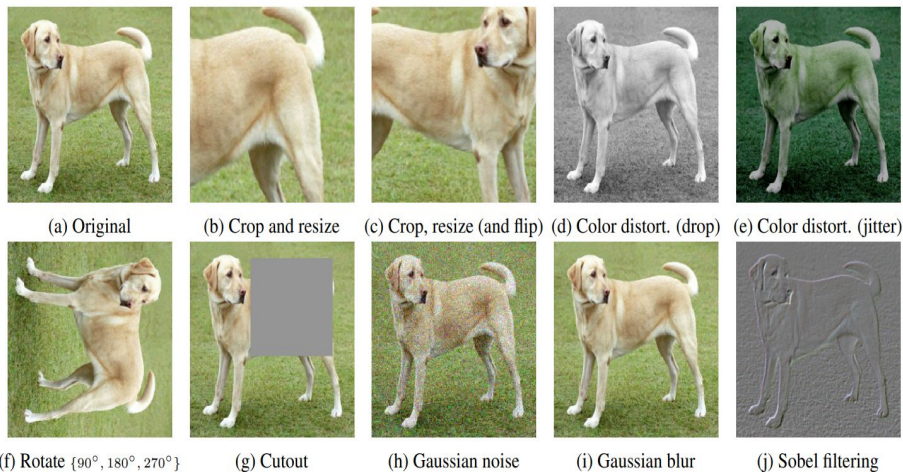
Contrastive representation learning

SimCLR: A simple framework for contrastive learning of visual representations



Contrastive representation learning

Contrastive-based methods aim to learn data representation by contrasting positive and negative samples



Contrastive learning in time series

MF-CLR (ICML'24): introduces a self-supervised framework designed to learn effective representations from multi-frequency time series

TimeDRL (ICDE'24): focuses on disentangled representation learning for multivariate time-series data

SoftCLT (ICLR'24): attempts to select more positive pairs by creating soft assignments using Dynamic Time Warping (DTW) as the distance metric.

SimMTM (Neurip'23): is a pre-training framework designed for masked time-series modeling

InfoTS (AAAI'23): maximizes agreement between representations of the same subseries through temporal augmentations

CoST (ICLR'22): employs a two-step approach to time series forecasting by learning disentangled seasonal trends

TS2Vec (AAAI'22): captures both global and temporal dependencies by contrasting time series across different scales and timesteps

TNC (ICLR'21): learns by contrasting data points within the same neighborhood against those from different neighborhoods

Contrastive learning in time series (Limitation)

- Augmentations and masking
- Dynamic Time Warping (DTW)
- Multiple models to capture granularity

Consequence

- Data distortion
- Scalability and speed

Looking at the temporal information

Similarity within nearby instances

- That consecutive instances in a sequence have the same class and labels would not change too often
- Often holds for time series, which have repeated labels in the temporal dimension

Dynamic contrastive learning for time series representation (under review)

Transformer and Self-Attention

- Discard Recurrent Neural Networks
- Pattern within the data (Sentence structure and context)
- Scalable

Dynamic contrastive learning

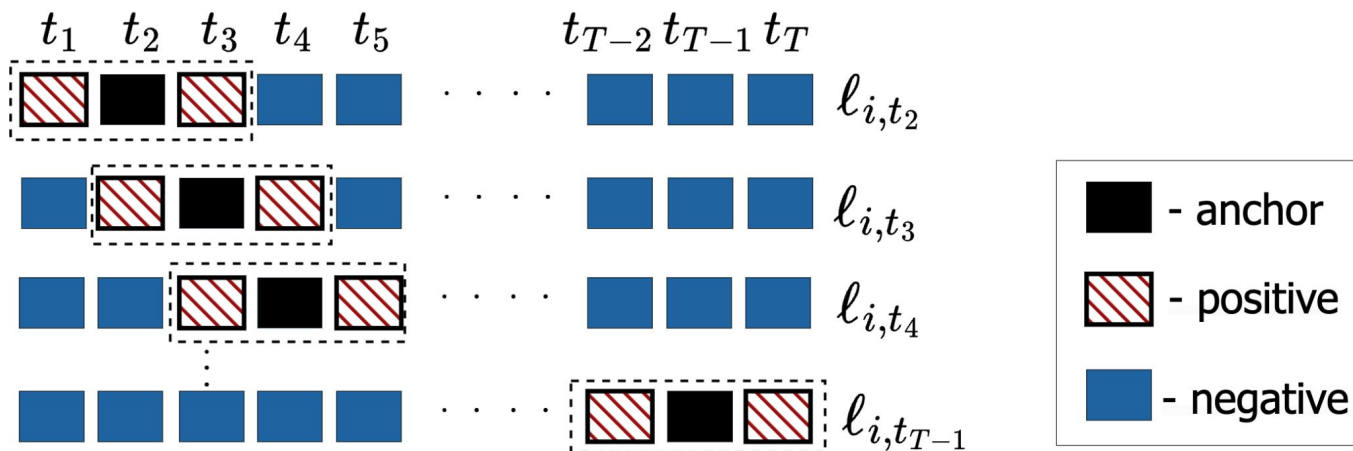
- Discard Augmentation
- Pattern within the data (Similarity within nearby instances)
- Scalable ?



source: image generated by Microsoft copilot

Dynamic contrastive learning (DynaCL)

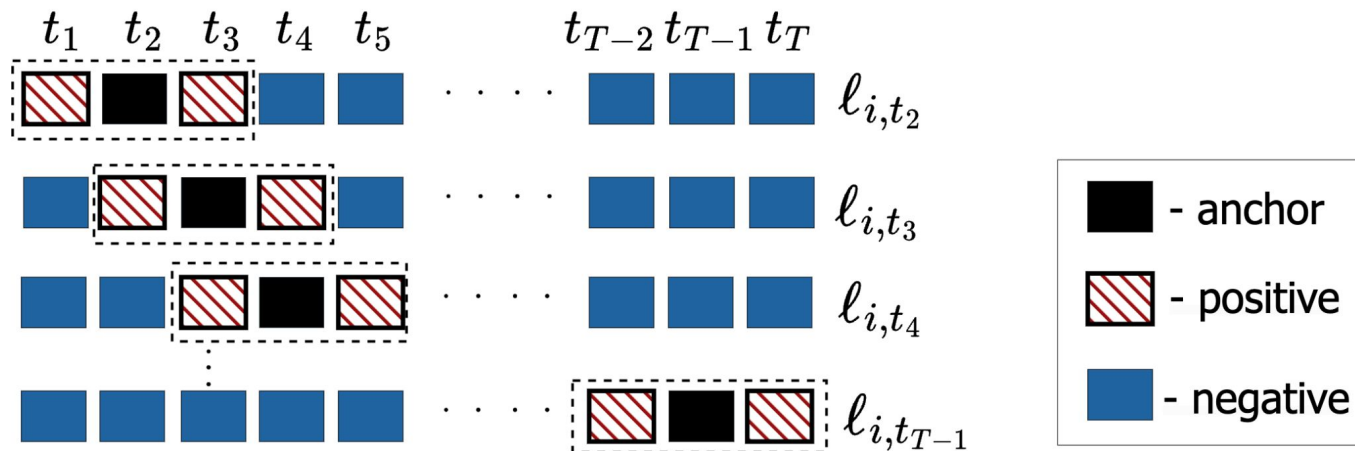
A naive way to select these multiple positives would involve slicing the temporal adjacent instances for all instances in a batch with a high computational complexity



Dynamic contrastive learning (DynaCL)

Adopts N-pair loss to dynamically treat all samples in a batch as positive and negative pairs at least once

or Contrast everything!



N-pair tuple loss

Positive pairs

$$\ell(i) = -\log \frac{\exp(\text{sim}(z_i, z_{i+1})/\tau)}{\sum_{k=1}^N \mathbf{1}_{[k \neq i, i+1]} \exp(\text{sim}(z_i, z_{k+1})/\tau)}$$

Negative pairs

	Instance 1	Instance 2	Instance 3	Instance 4	Instance 5	Instance 6
Instance 1		sim(1, 2)	sim(1, 3)	sim(1, 4)	sim(1, 5)	sim(1, 6)
Instance 2	sim(2, 1)		sim(2, 3)	sim(2, 4)	sim(2, 5)	sim(2, 6)
Instance 3	sim(3, 1)	sim(3, 2)		sim(3, 4)	sim(3, 5)	sim(3, 6)
Instance 4	sim(4, 1)	sim(4, 2)	sim(4, 3)		sim(4, 5)	sim(4, 6)
Instance 5	sim(5, 1)	sim(5, 2)	sim(5, 3)	sim(5, 4)		sim(5, 6)
Instance 6	sim(6, 1)	sim(6, 2)	sim(6, 3)	sim(6, 4)	sim(6, 5)	

create a N x N similarity matrix.

The lower diagonal elements represent positive pairs

The denominator is the sum of the column vectors minus the positives.

Use the numerator and denominator to compute the N-tuple loss

N-pair tuple loss - Multiple positive

Positive pairs

$$\ell(i) = -\log \frac{\exp(\text{sim}(z_i, z_{i-1})/\tau) + \exp(\text{sim}(z_i, z_{i+1})/\tau)}{\sum_{k=1}^N \mathbf{1}_{[k \neq i, i-2]} \exp(\text{sim}(z_i, z_{k+1})/\tau) + \sum_{l=1}^N \mathbf{1}_{[l \neq i, i-1]} \exp(\text{sim}(z_{i-1}, z_l)/\tau)}$$

Negative pairs

	Instance 1	Instance 2	Instance 3	Instance 4	Instance 5	Instance 6
Instance 1		sim(1, 2)	sim(1, 3)	sim(1, 4)	sim(1, 5)	sim(1, 6)
Instance 2	sim(2, 1)		sim(2, 3)	sim(2, 4)	sim(2, 5)	sim(2, 6)
Instance 3	sim(3, 1)	sim(3, 2)		sim(3, 4)	sim(3, 5)	sim(3, 6)
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Instance 5	sim(5, 1)	sim(5, 2)	sim(5, 3)	sim(5, 4)		sim(5, 6)
Instance 6	sim(6, 1)	sim(6, 2)	sim(6, 3)	sim(6, 4)	sim(6, 5)	

Similarity matrix N x N

Select the lower diagonal elements.

The positive pairs are the sum of the shifted left and right of the lower diagonal elements

The denominator is the sum of all elements in the similarity matrix

Subtract a combination of two lower diagonal slices from the sum to produce the denominator

Dynamic contrastive learning (DynaCL)

MP-Xent NT tuple loss per update (efficient training)

$$\ell(i, t) = -\log \frac{\exp(\text{sim}(z_{i,t}, z_{i,t-1})/\tau) + \exp(\text{sim}(z_{i,t}, z_{i,t+1})/\tau)}{\sum_{k=1}^T \mathbf{1}_{[k \neq t, t-1, t+1]} \exp(\text{sim}(z_{i,t}, z_{i,k+1})/\tau) + \sum_{l=1}^T \mathbf{1}_{[l \neq t, t-1]} \exp(\text{sim}(z_{i,t-1}, z_{i,l})/\tau)} \quad (2)$$

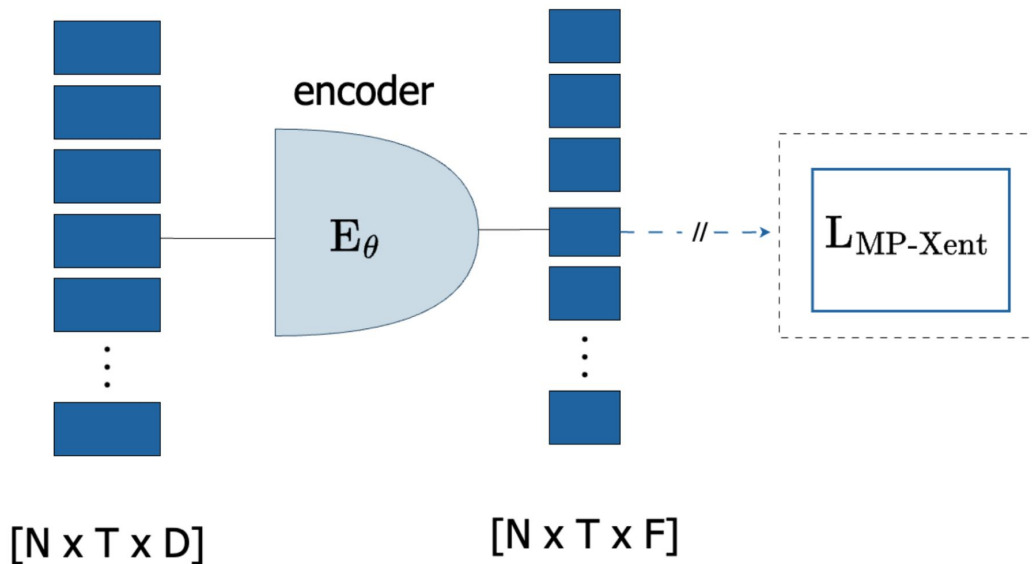
For the entire sequence length T and batch N , we have an NT tuple loss per update, making our training very efficient.

$$L_{MP-Xent} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \ell(i, t) \quad (3)$$

In our DynaCL model, we focus on optimizing only equation [3](#).

Dynamic contrastive learning (DynaCL)

MP-Xent NT tuple loss per update (efficient training)



Dataset distributions

Table 1. Summary of dataset distributions used in the classification experiments after applying STFT. An instance is a preprocessed block of TS. For the HARTH each instance is 1 second while for the SLEEP EEG and ECG an instance is 2 and 6 seconds respectively.

	# Instance	Sequence length	Dimension	Classes
HARTH	1,270,087	119	156	12
SLEEP EEG	478,600	200	90	5
ECG	255,200	200	1502	4

Linear evaluation with frozen backbone

Datasets	Models	Accuracy	F1 score	Precision	Recall	Time (s) ↓
HARTH	Random Init.	86.69±0.50	0.84±0.01	0.55±0.01	0.45±0.02	-
	TNC	89.03±0.03	0.88±0.00	0.70±0.02	0.55±0.01	13.86
	InfoTS	90.12±0.13	0.89±0.00	0.71±0.02	0.56±0.01	186.50
	CoST	88.68±0.41	0.88±0.00	0.64±0.01	0.56±0.00	12.61
	SimMTM	68.14±4.83	0.60±0.08	0.34±0.04	0.24±0.05	245.67
	TimeDRL	71.92±2.37	0.63±0.04	0.37±0.04	0.28±0.01	20.60
	TS2Vec	87.59±0.23	0.85±0.00	0.58±0.01	0.49±0.00	34.96
	TS2Vec + <i>SoftCLT</i>	88.07±0.13	0.86±0.00	0.62±0.03	0.50±0.01	37.65
	MF-CLR	77.07±5.38	0.73±0.07	0.52±0.04	0.37±0.05	2322.75
	DynaCL	90.35±0.05	0.89±0.00	0.71±0.02	0.59±0.00	6.12
SLEEP EEG	Random Init.	46.89±11.34	0.44±0.08	0.36±0.03	0.32±0.04	-
	TNC	55.13±0.50	0.54±0.01	0.33±0.01	0.36±0.02	10.14
	InfoTS	43.32±9.66	0.42±0.07	0.41±0.05	0.35±0.03	260.35
	CoST	56.63±2.56	0.55±0.00	0.38±0.02	0.35±0.04	11.89
	SimMTM	57.08±0.68	0.50±0.01	0.31±0.03	0.25±0.01	1739.66
	TimeDRL	57.30±2.22	0.48±0.02	0.36±0.09	0.24±0.02	13.20
	TS2Vec	51.18±1.12	0.50±0.00	0.29±0.01	0.35±0.02	34.42
	TS2Vec + <i>SoftCLT</i>	53.60±2.44	0.52±0.01	0.34±0.03	0.37±0.02	36.48
	MF-CLR	57.07±1.89	0.48±0.02	0.27±0.06	0.27±0.01	2318.26
	DynaCL	62.39±1.21	0.60±0.01	0.47±0.00	0.41±0.01	6.36
ECG	Random Init.	65.48±1.44	0.64±0.01	0.35±0.01	0.34±0.01	-
	TNC	75.86±2.00	0.75±0.02	0.39±0.00	0.38±0.01	4.86
	InfoTS	61.94±3.19	0.59±0.05	0.34±0.01	0.32±0.01	76.42
	CoST	69.37±9.55	0.67±0.13	0.38±0.02	0.36±0.04	11.27
	SimMTM	71.89±2.27	0.71±0.03	0.37±0.00	0.36±0.01	594.86
	TimeDRL	59.51±6.93	0.55±0.10	0.33±0.02	0.31±0.03	13.21
	TS2Vec	77.72±1.27	0.77±0.01	0.40±0.00	0.39±0.01	11.42
	TS2Vec + <i>SoftCLT</i>	76.37±3.88	0.76±0.04	0.39±0.02	0.39±0.02	16.70
	MF-CLR	52.69±2.29	0.45±0.06	0.31±0.01	0.28±0.01	3056.11
	DynaCL	80.18±3.80	0.81±0.03	0.41±0.01	0.41±0.02	4.47

DynaCL consistently achieves the highest accuracy

Lowest computational time, demonstrating its efficiency

Practical framework for activity recognition and physiological signal analysis

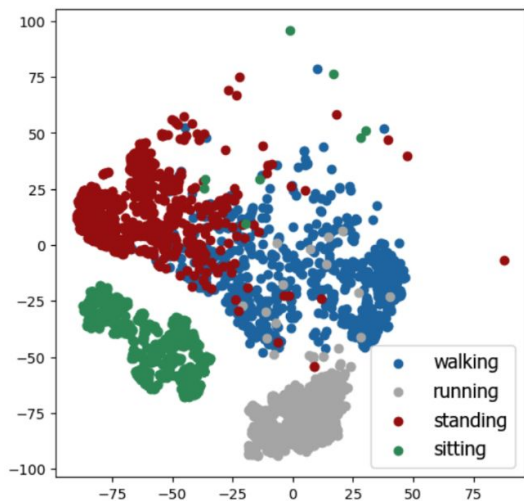
Linear evaluation with frozen backbone

	HARTH			SLEEPEEG			ECG		
	5%	10%	20%	5%	10%	20%	5%	10%	20%
TNC	84.96±2.05	86.78±0.89	87.94±0.23	54.71±2.57	54.02±2.23	52.70±3.10	63.87±4.18	71.65±6.03	73.32±7.33
InfoTS	87.39±0.92	88.16±0.64	88.69±0.64	40.34±11.25	37.80±11.51	43.50±9.26	52.30±8.30	55.55±6.97	57.16±6.39
CoST	86.84±0.70	87.59±0.43	88.24±0.28	55.13±2.83	53.96±3.52	55.70±2.99	62.68±3.51	65.90±2.78	66.90±4.79
SimMTM	59.35±2.13	60.07±1.37	61.06±1.37	57.39±1.58	57.72±1.61	56.93±2.04	56.63±5.93	60.90±5.62	65.86±4.54
TimeDRL	56.95±1.84	62.00±2.52	66.70±1.39	55.60±4.54	56.30±3.52	56.02±3.56	49.85±2.45	50.50±2.18	49.84±1.46
TS2Vec	79.18±2.27	83.98±0.70	85.36±0.51	57.80±1.39	55.05±2.84	51.85±2.11	58.46±8.32	71.11±6.96	74.23±6.88
TS2Vec + <i>SoftCLT</i>	83.08±1.47	85.02±1.02	86.36±0.63	56.35±1.02	54.24±2.09	54.65±2.34	63.24±9.57	69.30±8.73	75.32±7.36
MF-CLR	62.06±1.54	64.47±1.48	66.50±0.52	43.81±7.26	46.27±10.51	48.26±8.40	59.08±4.56	57.84±6.19	62.82±3.35
DynaCL	84.74±2.26	86.90±1.58	88.93±0.73	59.99±2.33	61.82±1.48	61.95±1.70	73.37±10.10	73.84±6.58	77.43±4.21

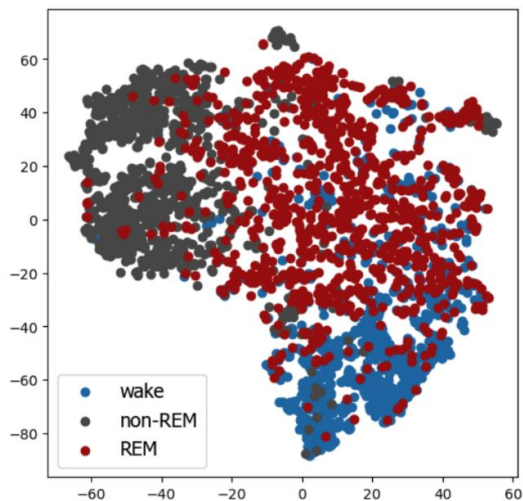
Visualization of learned representations

t-SNE plots of the learned embeddings

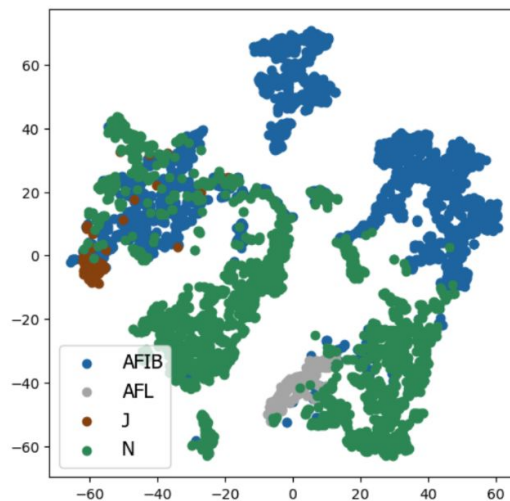
HARTH



SLEEP



ECG



Short and long horizon forecasting

Dataset	H	DynaCL		TS2Vec		TS2Vec + <i>SoftCLT</i>		CoST		MF-CLR	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh ₁	24	0.040	0.152	0.040	0.153	0.040	0.153	0.041	0.152	0.039	0.149
	48	0.063	0.193	0.063	0.193	0.063	0.192	0.066	0.196	0.062	0.189
	168	0.122	0.269	0.145	0.296	0.144	0.296	0.126	0.273	0.142	0.297
	336	0.143	0.296	0.164	0.321	0.164	0.321	0.150	0.305	0.165	0.323
	720	0.189	0.354	0.174	0.340	0.174	0.342	0.188	0.354	0.201	0.371
ETTh ₂	24	0.095	0.235	0.090	0.227	0.089	0.226	0.091	0.231	0.093	0.230
	48	0.130	0.279	0.126	0.275	0.125	0.274	0.127	0.277	0.133	0.279
	168	0.190	0.348	0.200	0.357	0.201	0.357	0.206	0.359	0.200	0.352
	336	0.196	0.357	0.205	0.365	0.206	0.366	0.206	0.363	0.206	0.363
	720	0.201	0.365	0.205	0.370	0.205	0.370	0.205	0.368	0.202	0.364
ETTm ₁	24	0.015	0.092	0.016	0.093	0.014	0.089	0.014	0.087	0.015	0.090
	48	0.028	0.124	0.034	0.137	0.027	0.123	0.027	0.121	0.028	0.125
	96	0.044	0.159	0.049	0.169	0.045	0.162	0.044	0.158	0.043	0.156
	288	0.086	0.225	0.099	0.252	0.097	0.238	0.092	0.233	0.094	0.234
	672	0.124	0.271	0.149	0.296	0.144	0.289	0.137	0.289	0.134	0.280
Weather	24	0.037	0.152	0.131	0.317	0.105	0.276	0.049	0.198	0.080	0.257
	48	0.024	0.114	0.094	0.226	0.077	0.202	0.034	0.153	0.049	0.181
	168	0.012	0.082	0.034	0.120	0.030	0.118	0.013	0.080	0.017	0.086
	336	0.009	0.069	0.021	0.097	0.019	0.094	0.011	0.074	0.012	0.074
	720	0.011	0.083	0.015	0.086	0.014	0.084	0.009	0.070	0.011	0.075
Avg.		0.088	0.211	0.103	0.235	0.099	0.229	0.092	0.217	0.096	0.224

Timestamp level resolution

Overall DynaCL achieves the lowest prediction MSE

Lowest computational time

	TIME (s)				
	DynaCL	TS2Vec	TS2Vec + <i>SoftCLT</i>	CoST	MF-CLR
ETTh ₁	4.62	16.12	16.93	6.70	239.33
ETTh ₂	3.02	15.63	18.50	6.69	216.32
ETTm ₁	14.43	65.37	66.11	15.80	2604.35
Weather	18.05	63.95	68.12	14.89	2332.83
Total	40.12	161.07	169.66	44.08	5392.83

Ablation study on Weather and ECG

Model Variants	<i>forecasting</i>	
	MSE	MAE
Single positive w/o negatives	0.014	0.095
Single positive + negatives	0.012	0.085
Multiple positive + negatives (MP-Xent)	0.011 (-21.4%)	0.083 (-12.6%)

	<i>classification</i>	
	Accuracy	F1 Score
Single positive w/o negatives	62.41	0.58
Single positive + negatives	71.69	0.71
Multiple positive + negatives (MP-Xent)	80.18 (+28.4%)	0.81 (+39.7%)

What did we learn?

Key Findings

- Focusing on temporal information rather than augmentation might be the way forward
- Demonstrates the ability to learn semantically meaningful representations
- Outperforms previous time series representation learning methods in downstream classification and forecasting tasks
- Learns timestamp-level representation, as underscored by its performance in time series forecasting
- Delivers exceptional performance in downstream classification and forecasting tasks with the shortest training time

Questions?